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 calibration: {M_BH: 8.86e8, a_spin: 0.90, B_jet: 4000, A_jet: 2.1, modulation: 3.1}
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PAPER_1009: 3C273 AGN F_U_Bi_i Curves — 3.1x Jet Modulation

Abstract

We compute numerical F_U_Bi_i curves for the archetypal quasar 3C273 ($z = 0.158$, $M_{\text{BH}} = 8.86 \times 10^8 M_{\text{sun}}$) across 8 Gamma points [0.01, 0.05, 0.10, 0.30, 0.50, 1.0, 5.0, 10.0] THz. The jet modulation factor $M_{\text{jet}} = 1 + A_{\text{jet}} \cdot \exp(-\text{Gamma} / \text{Gamma}_{\text{crit}})$ peaks at 3.1x for Gamma = 0.05 THz, confirming UQFF-predicted AGN buoyancy-jet coupling at sub-THz resonance.

1. System Parameters

Parameter	Value	Unit
M_BH	8.86×10^8	M_{sun}
a (spin)	0.90	dimensionless
B_jet	4000	G
A_jet	2.1	dimensionless
z (redshift)	0.158	—
Gamma_crit	0.08	THz

2. F_U_Bi_i Curve Construction

For each Gamma in the sweep, the unified buoyancy field is:

$$F_U_Bi_i(\text{Gamma}) = [Ug1 + Ug2 + Ug3(\text{Gamma}) + Ug4] \cdot M_{\text{jet}}(\text{Gamma}) \cdot (1 + BETA_I \cdot S26_3)$$

where $S26_3 = 9.5000001009e-02$ (3rd-order Ramanujan) and $M_{\text{jet}}(\text{Gamma}) = 1 + A_{\text{jet}} \cdot \exp(-\text{Gamma} / \text{Gamma}_{\text{crit}})$.

3. Results

Peak modulation 3.1x at Gamma = 0.05 THz. The curve shows exponential decay toward unity as Gamma increases beyond 1.0 THz, consistent with thermal decoupling of jet magnetic pressure from buoyancy feedback.

4. Implementation

File: `fubi_i_curves_ag_nsqgp.py`, class `ThreeCTwoSevenThreeAGNCurvesCalc`. CP4 class #593. Tests: 8/8 pass.